JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 96, NO. B6, PAGES 9887–9894, JUNE 10, 1991

THE ELKHORN MOUNTAINS REVISITED: NEW DATA FOR THE LATE CRETACEOUS PALEOMAGNETIC FIELD OF NORTH AMERICA

J. F. Diehl

Department of Geological Engineering, Geology, and Geophysics
Michigan Technological University, Houghton

Abstract. The Late Cretaceous portion of North America's apparent polar wander (APW) path is poorly known due to a dearth of suitable post-85-Ma reference poles. A paleomagnetic re-investigation of Elkhorn Mountains Volcanics (80 Ma) was initiated to fill this void and to determine the degree to which the onset of an apparent rapid episode of APW during this time interval correlates with known Late Cretaceous tectonic events. A total of 249 paleomagnetic samples (30 sites) were collected from three separate areas within the Elkhorn Mountains. Using both alternating field and thermal demagnetization, a characteristic direction of magnetization was isolate at 15 sites ($a_{95} < 15^{o}$) which passes a fold test at the 95% confidence level. The mean pole calculated from site virtual geomagnetic poles is located at 80.3°N, 189.5°E (N = 15; A_{95} = 9.6°; k = 16.8). Interestingly, the Elkhorn Mountains pole is not concordant with the Cretaceous stillstand poles but is nearly coincident with the Paleocene reference pole and the recently published pole for the Adel Mountains dated at 76 Ma. These new data suggest the onset of APW which brought the pole from its Cretaceous position to near its early Tertiary position occurred prior to 80 Ma and the transition was exceedingly rapid. The onset of this rapid APW now appears to coincide more closely in time with many of the significant Late Cretaceous tectonic events, e.g., the Laramide orogeny. However, since the Elkhorn Mountains Volcanics are floored by a major east-directed thrust and have undergone crustal scale deformation, the data should be viewed with caution until more results are available for this time interval.

Introduction

Studies of Cretaceous and early Tertiary rocks from the North American craton have established that (1) little if any apparent polar wander (APW) occurred during much of the Cretaceous (130-85 Ma), (2) a significant episode of APW (~10°-13°) occurred during the Late Cretaceous, and (3) slow apparent polar motion occurred thereafter. However, the Late Cretaceous episode of APW is not well documented due to a lack of suitable reference poles younger than 85 Ma. It is imperative to delineate this segment of North America's APW path so that proper reference poles exist for this time interval. In addition, it is important to determine the degree to which the onset of this APW episode correlates with Late Cretaceous/early Tertiary tectonic events such as the beginning of the Laramide orogeny and increased absolute motion of the North American plate both in the hotspot reference frame and with respect to the Farallon plate.

To further add to the Late Cretaceous database, a re-investigation of the Elkhorn Mountains Volcanics was initiated. The original study by Hanna [1967] reported a paleomagnetic pole concordant with other Cretaceous poles and was used in many earlier calculations of the Cretaceous reference field. The pole has now become suspect and has not been used in most recent compilations of Cretaceous data [e.g., Globerman and Irving, 1988].

Geologic Setting

The Elkhorn Mountains Volcanics are located in southwestern Montana near the margins of the Boulder Batholith and west of the leading edge of Montana's fold and thrust belt. Named by Klepper et al. [1957], these rocks are the erosional remnants of a plateau-like volcanic field which may have covered 26,000 km^2 with a thickness of 2-3 km [Klepper et al., 1957; Smedes, 1966]. The volcanic sequence has been divided into three informal members. The lower member consists predominantly of volcanic flows (andesite and rhyodacite), autobrecciated lavas, volcanic sediments, and some thin quartz latite ash flows. The middle member consists of densely welded rhyolitic ash flow sheets and interbedded ashfall sediments, while the upper member consists largely of water-laid andesitic volcanic sediments. These lithostratigraphic units vary in thickness from area to area and crop out discontinuously, making regional stratigraphic correlations ambiguous.

Based on stratigraphic and paleontologic data [Klepper et al., 1957; Smedes, 1966; Robinson et al., 1968], the Elkhorn Mountains Volcanics are early Campanian in age. K-Ar dates (using the new International Union of Geological Sciences constants) from the lower member of the Elkhorn Mountains Volcanics (80.8 ± 2.4 Ma and 79.6 ± 2.4 Ma) reinforce this age [Tilling et al., 1968]. These dates together with dates from the oldest intrusive phases of the Boulder Batholith which cut the volcanics indicate volcanism commenced at approximately 80 Ma and lasted only a few million years [Lipman, 1984].

Field and Laboratory Procedures

Samples were collected using portable drilling equipment and a solar compass for orientation.

Copyright 1991 by the American Geophysical Union.

Paper number 91JB00959.
0148-0227/91/91JB-00959$05.00

Structural attitudes at the sites were determined from flow contacts, interbedded sediments, eutaxitic structures found in the welded-tuffs, or a combination of the aforementioned. Additional structural information was taken from the many geological maps published on the area [Klepper et al., 1957, 1971; Weeks, 1974; Streeter, 1983]. Figure 1 shows the location of the three sampling areas for this study; all are east of the Boulder Batholith. They included the southeastern foothills of Bull Mountain originally sampled by Hanna [1967], the southern Elkhorn Mountains as shown in Plate 1 by Klepper et al. [1957], herein referred to as the South Elkhorn section, and the East flank of the Elkhorn Mountains [Klepper et al., 1971] between the latitudes defined by the towns of Winston and Townsend, Montana.

A total of 109 paleomagnetic cores from 13 sites were collected from the Bull Mountain section. Six sites were from welded-tuff units, two from lava flows, two from intrusive units, and three from volcaniclastic units. These units most likely represent the lower and middle members mapped in other areas of the Elkhorn Mountains. The volcanic sequence is over 1 km thick [Hanna, 1967] with a fairly uniform strike to the NW and a dip of 50°-65° to the NE. In the southern Elkhorn Mountains, a total of 100 oriented samples were collected from 12 sites. The majority of units sampled belong to the lower member; four of the sites are from andesite flows, two from welded-tuffs, and six from volcaniclastic units. The units dip between 15° and 22°. Along the eastern flank of the Elkhorn Mountains only five sites (40 samples) were collected due to a lack of good exposure. The units sampled represent both the lower and middle members. Two sites were from welded-tuffs, one was from a volcaniclastic unit, and two were located in hypabyssal intrusions. Dips were between 15° and 25°. All but two sites from these three areas were collected from natural exposures, and because of the limited amount of drillable exposure many of these were on or near ridge crests.

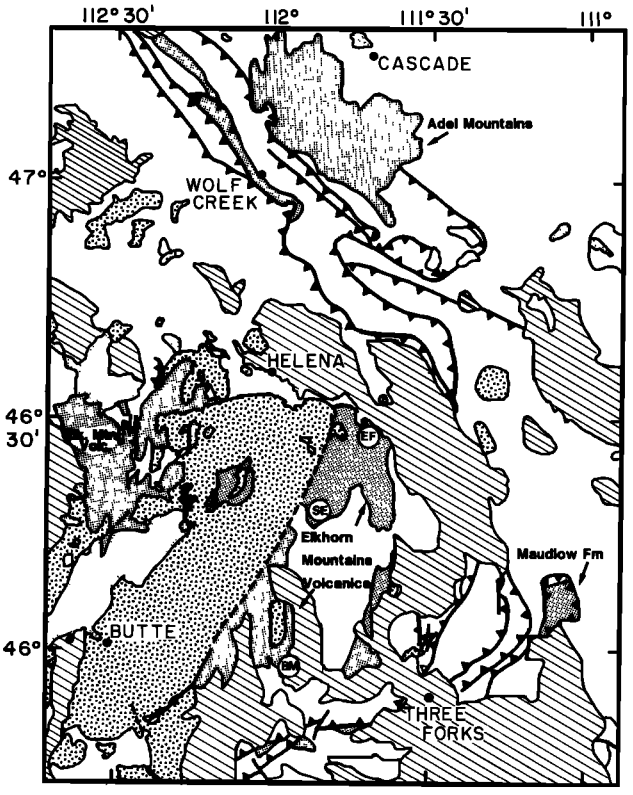


Fig. 1. Generalized geologic map of southwestern/west-central Montana modified from Robinson et al. [1968]. Dark stippling, Late Cretaceous volcanic rocks; triangular stippling, Late Cretaceous plutonic rocks; light stippling, Tertiary volcanic rocks; diagonal lines, Cenozoic basin deposits. BM, SE, and EF are the Bull Mountain, South Elkhorn, and East flank sampling localities, respectively. Arcuate pattern of thrust faults defines the Helena salient.

The remanent magnetization (RM) of individual specimens cut from each paleomagnetic core was measured using either the spinner magnetometer housed in the paleomagnetics laboratory at Western Washington University or at Michigan Technological University. Pilot specimens from each site were selected for alternating field (AF) and thermal demagnetization experiments. For AF demagnetization, specimens were stepwise demagnetized using increments of either 5 or 10 mT up to peak inductions of 80-100 mT using either a three-axis tumbling or a single-axis nontumbling demagnetizer. Thermal demagnetization was performed in air using steps between 15° and 100°C to temperatures of either 590° or 690°C using a homemade noninductively wound furnace. Specimens were held at demagnetization temperature for 30 min and then cooled in a magnetically shielded space where the ambient field was less than 10 nT. The demagnetization method most effective in isolating the pilot specimens' characteristic remanence (ChRM) from secondary components was used on the remaining specimens from that site using a minimum of at least four demagnetization steps.

Data were plotted on standard orthogonal demagnetization diagrams [Zijderveld, 1967] and visually inspected. The ChRM of each specimen was determined by applying principal component analysis to the vector endpoints which defined a linear trend to the origin. Where two or more components of magnetization had overlapping coercivity spectra, making it difficult to directly resolve the specimen's ChRM, remagnetization circles (demagnetization planes) were fit to the data. Site-mean directions were determined either by using the statistical methods of Fisher [1953], for sites where the ChRM of each specimen was successfully isolated, or by using the maximum likelihood analysis method of McFadden and McElhinny [1988] that combines remagnetization circles and ChRMs so standard Fisher [1953] statistics apply. Specimens in which a ChRM could not be identified or had directions markedly different from those of other specimens from the same site were not used in calculating a mean direction for the site. In the latter case, many of these specimens came from sites located near ridge crests, had high RM intensities, and were probably lightning struck.

Seventeen sites yielded site-mean directions with circles of 95% confidence (a_{95}) less than an arbitrary cutoff value of 15°; the pertinent paleomagnetic data for

21562202b, 1991, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/91JB00959 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

these 17 sites are listed in Table 1. The remaining 13 sites had either site-means with a_{95}'s greater than 35° or too few samples to be considered reliable estimates of the Late Cretaceous geomagnetic field. These sites were not used in the final statistical analysis and are not listed in Table 1.

Eleven of the 17 reliable sites come from the Bull Mountain section. The orthogonal demagnetization diagrams shown in Figures 2a, 2b, and 2c are typical of specimens from these sites. After the initial removal of a secondary magnetization, a univectorial decay of the RM is achieved. Both thermal and AF demagnetization isolate the same ChRM in these sites. Only 10 specimens required remagnetization circle analysis to recover the ChRM of the sample. In some instances, the RM is entirely unblocked by 580°C, indicating that magnetite is the remanence carrier, while in other samples the RM is not entirely unblocked until 690°C, indicating hematite contributes to the remanence. Where both a low unblocking temperature component (magnetite) and a high unblocking temperature component (hematite) exist in the same specimen, they carry the same direction. For the two Bull Mountain sites not yielding reliable site-mean directions, most of the specimens displayed erratic demagnetization behavior in which neither a linear segment nor a great circle path could be statistically identified.

Only four sites from the South Elkhorn section and two sites from the East flank of the Elkhorn Mountains produced reliable site-mean directions. Vector endpoint diagrams (Figures 2d, and 2e) from these sites are similar to those from the Bull Mountain section with 11 specimens requiring great circle analysis to define the ChRM. Of the 11 sites not yielding reliable directions from these two areas, four sites were seriously affected by lightning strikes and yielded too few specimens to give statistically meaningful results. Many specimens had exceedingly high intensities and displayed a classic isothermal remanent decay curve characteristic of rocks carrying a lightning-induced magnetization [Cox, 1961; Graham, 1961; Globerman and Irving, 1988]. All were collected from near ridge crests or prominent topographic features. Two sites were from welded-tuff units which had highly elongated fiamme. Individual specimens showed univectorial decay of the RM, but each vector had a different direction. Rosenbaum [1986] and Hagstrum and Gans [1989] have presented evidence to show that this type of directional dispersion in welded-

TABLE 1. Site-Mean Directions From the Elkhorn Mountains Volcanics

Site	Slat	Slong	N/N_O	D	I	a_{95}	k	DipAz/DipAn	D_C	I_C	Plat	Plong
					Bull Mountain Section							
em1	45.93	111.99	7/7	259.3	42.7	5.2	135	55/50	341.3	71.5	74.5	206.1
em2	45.93	111.99	7/7	74.6	-29.2	8.7	51	55/50	117.1	-70.6	50.4	194.7
em3	45.94	111.99	12/16	72.6	-33.8	10.9	17	55/55	145.1	-75.5	64.2	210.8
em4	45.93	111.98	8/8	96.3	-40.0	4.2	177	55/65	178.9	-52.5	77.1	72.3
em6	45.93	111.98	6/8	274.4	50.8	8.2	74					
em7	45.93	111.98	8/8	68.6	-28.5	6.4	90	40/60	133.8	-64.9	58.6	176.8
em8	45.93	111.99	7/8	79.3	-27.2	10.2	36	55/60	143.5	-68.5	65.6	185.1
em9	45.94	111.98	8/8	85.5	-35.9	10.8	27	55/60	166.1	-63.8	80.3	160.5
em11	45.94	111.98	8/8	67.4	-33.6	2.5	500	45/65	164.1	-68.7	77.9	194.3
em12	45.94	111.98	8/8	80.8	-54.0	3.1	326	50/65	199.9	-53.1	70.5	10.0
em13	45.94	111.98	5/7	83.1	-32.8	3.5	481	55/65	169.2	-64.3	82.5	163.2
					South Elkhorn Section							
em14	46.26	111.91	8/8	133.3	-67.3	9.0	44	75/14	168.6	-70.7	78.7	213.0
em16	46.26	111.90	7/8	186.8	-66.0	9.8	41	75/14	208.2	-58.2	68.1	347.5
em19	46.28	111.88	7/9	172.8	-53.1	5.4	128	60/16	187.7	-44.8	69.2	47.9
em23	46.46	111.74	6/8	352.8	77.3	12.5	37	295/15	322.6	65.8	64.8	175.6
					East Flank Section							
em25	46.32	111.81	7/8	309.8	-79.7	2.4	643	339/21	181.7	-77.0	71.0	250.4
em30	46.30	111.82	7/8	322.6	62.1	7.4	67					

Slat, latitude (°N) of sampling site; Slong, longitude (°W) of sampling site; N/N_O, ratio of the number of samples collected to those used in determining site-mean direction; D, I, D_C, I_C, declination and inclination before and after tilt correction; a_{95}, and k, Fisher [1953] statistical parameters; DipAz/DipAn, dip direction and angle; Plat, Plong, latitude (°N) and longitude (°E) of site mean. D_C, I_C from South Elkhorn and East flank sites are plunge-corrected directions.

21562202b, 1991, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/91JB00959 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

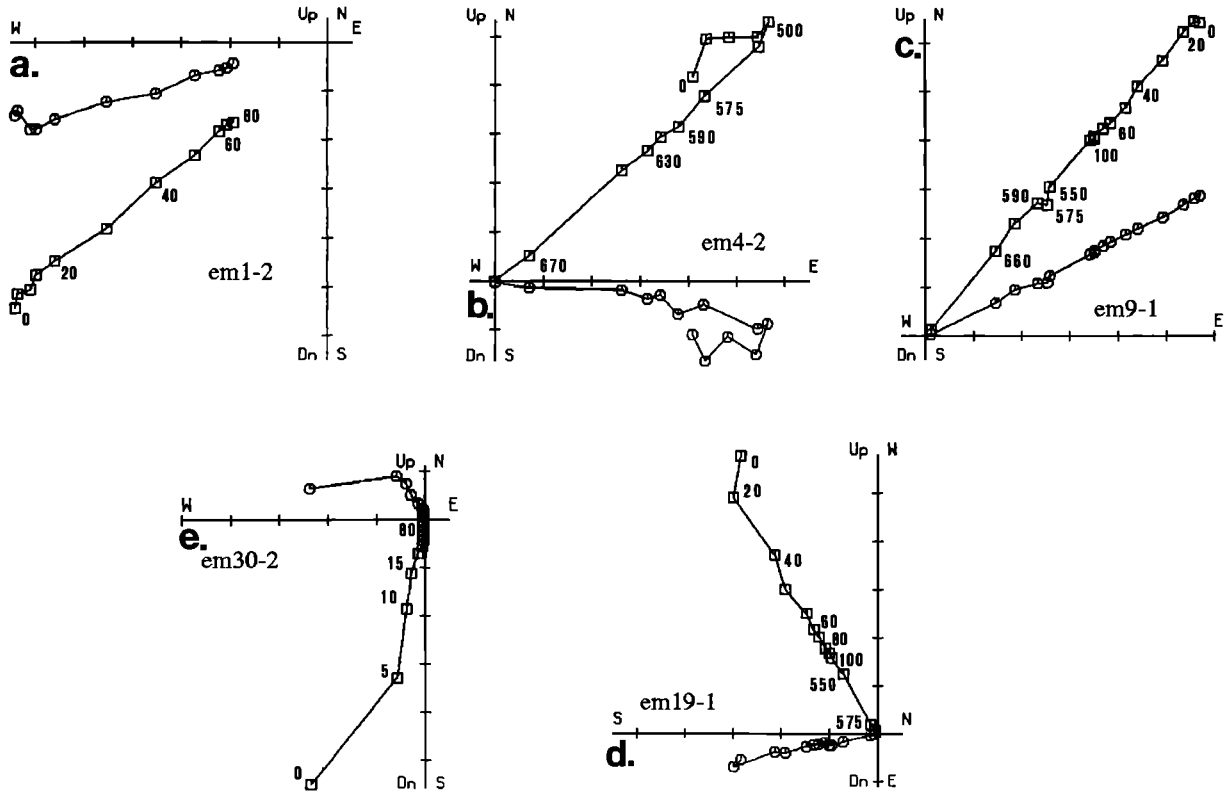


Fig. 2. Representative orthogonal demagnetization diagrams for the Elkhorn Mountains Volcanics. Open circles are projections of the remanent magnetization vector in geographic coordinates onto the horizontal plane; open squares are projections onto the vertical plane. Figures 2a and 2e are alternating field (mT) demagnetization, Figure 2b is thermal (°C) only, and Figures 2c and 2d are alternating field followed by thermal. Horizontal and vertical axes divisions are (a) 2.3 mA/m; (b) 1.4×10^2 mA/m; (c) 5.0×10^1 mA/m; (d) 8.6×10^2 mA/m; (e) 8.0×10^1 mA/m.

tuffs could be caused by sub-Curie temperature rheomorphic flow, and it is surmised that this is the cause of the dispersion. Four of the remaining five sites were from volcaniclastics. Neither AF nor thermal demagnetization was very successful in isolating a coherent direction for these sites. Some specimens displayed univectorial decay of the RM upon demagnetization, but the direction of the remanent vector varied from specimen to specimen. It is possible these volcaniclastics were emplaced at temperatures considerably below the Curie temperatures of the individual clasts composing the specimen. The remaining site was located in a weakly magnetized intrusion (RM intensities less than 10^{-3} A/m) and the isolated RMs were highly scattered.

Discussion

Figure 3 shows the distribution of the site-mean directions before and after tilt correction. Tilt corrections were made by rotating the unit about its present geologic strike. Not shown in Figure 3 are the results from sites em6 and em30. Both these sites were located in shallow-level intrusions that may have been emplaced before, after, or during folding [Smedes, 1966; Robinson et al., 1968]. Since the time of emplacement of these intrusions with respect to folding of the volcanic pile is uncertain and cannot be determined by this study due to a lack of acceptable sites located in intrusions, these sites were not used in calculating a mean direction for the Elkhorn Mountains Volcanics and are therefore not shown in Figure 3. As seen in Figure 3, the site-mean directions before tilt correction form two distinct groups. The 11 sites from the Bull Mountain section form a tight cluster with a easterly direction, while the four acceptable sites from the South Elkhorn and East flank sections form a more scattered group with a southerly direction. After correction for tilt, the site means form a single closely clustered group. The best estimate of Fisher's [1953] precision parameter, k, increases by a factor of 4.6 after tilt correction, an increase that is significant at the 95% confidence level. The fold test is therefore positive, indicating the ChRM isolated in these rocks was acquired before folding and probably represents a magnetization acquired at the time these rocks cooled and/or were deposited.

The Elkhorn Mountains Volcanics, however, are exposed in broad, open folds that generally trend north to northeast, with map patterns indicating some amount of plunge. Not correcting for plunge could introduce a significant error in the mean declination determined from these rocks [MacDonald, 1980]. To determine the degree to which fold axes deviate from the horizontal for the three areas, S-pole (or Pi) diagrams were constructed

21562202b, 1991, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/91JB00959 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

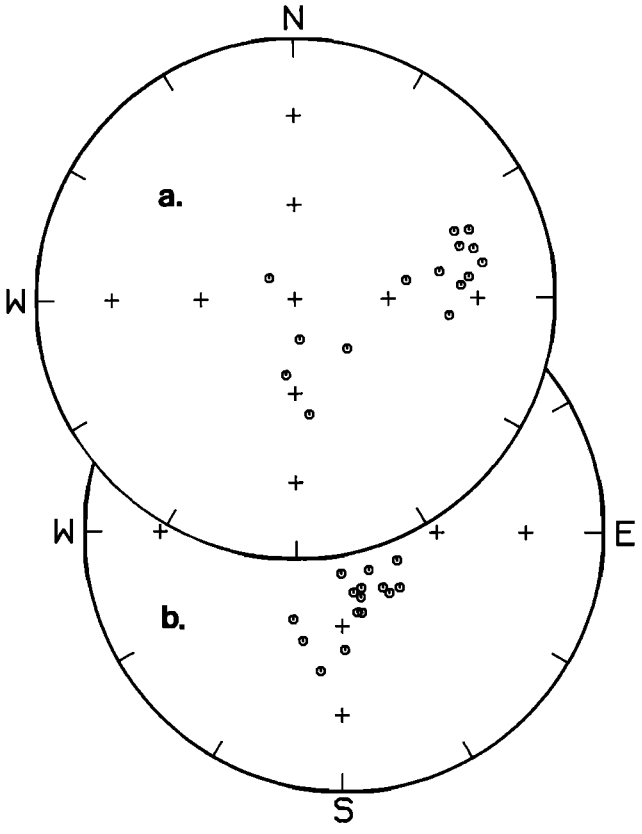


Fig. 3. Upper hemisphere equal area projection showing the distribution of site-mean directions (a) before (k = 7.4) and (b) after (k = 34.2) structural correction. Site means were tilt-corrected by rotation about a horizontal axis perpendicular to the dip direction given in Table 1. Normal-polarity directions inverted to reverse directions.

using structural attitudes measured from geologic maps and field-determined data. Two of the areas sampled (the South Elkhorn and East flank sections) had well-defined great circle distributions that allowed unambiguous determination of the fold axes. For units collected from the South Elkhorn section, the folds axes plunged 10° along a trend of N22°E, and for the sites taken from the East flank section, fold axes plunged 7° along a trend N27°W. The tilt-corrected directions listed in Table 1 have had the effect of plunge removed. The determination of the fold axis for the Bull Mountain section was more problematical. The section forms only one limb of a fold cut off by a normal fault [Streeter, 1983]. As a result, the volcanics have a fairly uniform strike over their outcrop extent (5-8 km of strike length). The poles to bedding display a tight (Θ_{67} = 2.6°) uniform distribution, making it impossible to define a great circle and thereby the fold axis with any degree of confidence. Hanna [1967] reported that no fold axes in the region around the Bull Mountain section plunged more than 5°, an amount insufficient to cause an error in declination by unfolding about a horizontal strike line. Results from the South Elkhorn and East flank sections support Hanna's [1967] contention that the amount of plunge is small. Interestingly, the fold test that gives the tightest clustering of directions (i.e., largest kappa) is that in which the South Elkhorn and East flank sites are corrected for plunge and a horizontal fold axis is assumed for the Bull Mountain section. Therefore, sites from the Bull Mountain section were tilt-corrected assuming a horizontal fold axis, while the South Elkhorn and East flank sites were also plunge-corrected.

The mean direction after inverting the directions of the normal-polarity sites is D = 168.8°, I = -66.7° (N = 15; a_{95} = 6.6°; k = 34.6). The mean direction determined from the normal-polarity sites is 8.2° of arc from being antipodal to the mean from the reversed-polarity sites, a difference that is not significant at 95% confidence level. Since there are only two normal-polarity sites, this is not a robust reversals test, but the data are suggestive that the ChRM directions determined from the Elkhorn Mountains Volcanics are not contaminated by secondary components of magnetization. In addition, 7 of the 15 sites are from welded-tuff units. Hagstrum and Gans [1989] have pointed out the potential for error in using eutaxitic structures as indicators of paleohorizontal. The mean direction determined from the seven welded-tuff sites (em2, em3, em7, em11, em12, em13, em23) differs by 8.9° of arc from the mean determined from the other eight sites, again a difference not significant at the 95% confidence level. In fact, the angular dispersions of directions from the two data sets are nearly identical (12.3° vs 12.9°), indicating that for this study data from eutaxitic structure are as reliable as those from flow contacts and interbedded sediments in determining paleohorizontal.

The corresponding paleomagnetic pole calculated from the site-mean virtual geomagnetic poles (VGPs) is located at 80.3°N, 189.5°E (A_{95} = 9.6°; k = 16.8). The angular dispersion of the 15 VGPs is 19.2° with upper and lower 95% confidence limits of 25.4° and 10.9°, respectively [Cox, 1969]. Although this value is slightly higher than the 17.1° predicted by Model G of McFadden et al. [1988] for the paleolatitude of the Elkhorn Mountains, it is very similar to the value of 18.9° for the Late Cretaceous (76 Ma) volcanic rocks from the Adel Mountains [Gunderson and Sheriff, 1991]. Whereas the original Elkhorn Mountains pole (69°N, 189°E) is very similar to other mid-Cretaceous paleomagnetic poles [Hanna, 1967], this new paleomagnetic pole is not. Gunderson and Sheriff [1991], however, have noted that the original pole cannot be calculated from the site-mean directions and structural corrections presented by Hanna [1967]. Therefore, the reported pole position is incorrect and cannot be compared to results from this study. The updated Elkhorn Mountains pole, as seen in Figure 4, is nearly coincident (1.3° of arc separation) with the Paleocene reference pole of Diehl et al. [1983] and is clearly separated (9.4° of arc separation) from the preferred mid-Cretaceous reference pole of Globerman and Irving [1988].

In the last several years, two studies have added to the Late Cretaceous database for North America. The paleomagnetic poles from the Maudlow Formation of southwestern Montana [Swenson and McWilliams, 1989] and the Adel Mountain volcanics of west-central

21562202b, 1991, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/91JB00959 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

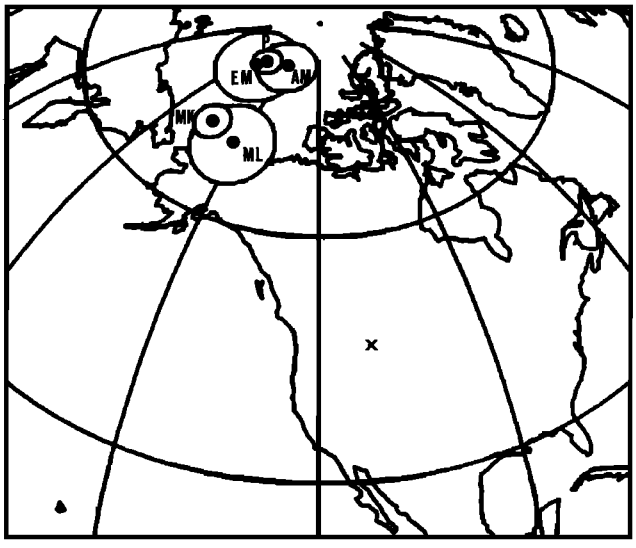


Fig. 4. Late Cretaceous paleomagnetic poles and their standard error circles (P = 0.37) from southwestern and west-central Montana. EM, Elkhorn Mountains Volcanics (this study); ML, Maudlow Formation [Swenson and McWilliams, 1989]; AM, Adel Mountains [Gunderson and Sheriff, 1991]. Also shown are the preferred mid-Cretaceous reference pole (MK) of Globerman and Irving [1988] and the Paleocene reference pole (P) of Diehl et al. [1983]. X is the sampling locality of this study.

Montana [Gunderson and Sheriff, 1991] are shown in Figure 4 along with the Elkhorn Mountains pole and the mid-Cretaceous and Paleocene reference poles. The Maudlow Formation is latest Santonian to latest Campanian in age and is thought to be the time equivalent of the Elkhorn Mountains Volcanics [Skipp and Peterson, 1965; Skipp and McGrew, 1977]. The Maudlow pole, located at 69.8°N, 207.8°E (A_{95} = 9.8°), has a magnetization that predates folding and is not statistically different from the mid-Cretaceous reference pole. The rocks of the Adel Mountain volcanic field have a nominal age of 76 Ma and overlie the Two Medicine Formation, which is stratigraphically equivalent to the Elkhorn Mountains Volcanics [Schmidt, 1972]. The Adel Mountain pole, located at 82.2°N, 209.9°E (A_{95} = 6.8°), passes a fold test and lies only 2.5° from the Paleocene reference pole. More importantly, all sites collected from the Adel Mountain volcanic field by Gunderson and Sheriff [1991] were located east of the leading edge of the Rocky Mountain fold and thrust belt and therefore have not undergone major crustal deformation since their emplacement.

Based on their results from the Adel Mountains, Gunderson and Sheriff [1991] have argued that the episode of APW which brought the pole from its mid-Cretaceous position to near its early Tertiary position occurred between 85 Ma and 76 Ma. Since the Elkhorn pole and the Adel pole are virtually identical, the Elkhorn data support Gunderson and Sheriff's [1991] hypothesis and suggest that the onset of APW occurred prior to 80 Ma and that it was exceedingly rapid (~2°/m.y.). The onset of this rapid APW, the start-up of the Laramide orogeny, and an increase in the absolute motion of the North American plate in the hotspot reference [Engebretson et al., 1985] all appear to coincide. Previous investigators have noted this close association [Diehl et al., 1980; Vugteveen et al., 1981; Gunderson and Sheriff, 1991], but the time window for the onset of this APW was much larger or was offset to a younger age. Unfortunately, data from the Maudlow Formation do not necessarily support a pre-80-Ma onset of APW as its pole is coincident with the mid-Cretaceous reference pole. Both the Elkhorn Mountains Volcanics and the Maudlow Formation lie in the Helena salient of Montana's fold and thrust belt (see Figure 1) and are underlain by major east directed thrusts [Ruppel et al., 1981]. Eldridge and Van der Voo [1988] have shown that buttressing effects of the Rocky Mountain foreland have caused local rotations of thrust sheets along the margin of the Helena salient. Local rotations of the two areas could explain the discrepancy between the two poles, but the Elkhorn Mountains Volcanics are quite outboard from the margins of the thrust sheets. In contrast, the sampling sites for the Maudlow Formation are within 15 km of Eldridge and Van der Voo's [1988] sampling site #12 which indicates that the Lower Cretaceous Kootenai Formation at this location has undergone 24° of clockwise rotation. Thus it is quite likely that the Maudlow Formation (and its paleomagnetic pole) have been affected by vertical-axis rotation. Alternatively, the magnetization ages of the Maudlow and Elkhorn poles may not be the same and both represent valid positions of the paleomagnetic pole that bracket this interval of rapid APW. However, it is unlikely the two poles have significantly different magnetization times even though the Maudlow Formation has slightly older K-Ar ages [Robinson and Marvin, 1967; Skipp and McGrew, 1977]. Data used to compute both poles are dominated by reversed-polarity magnetizations. Given the stratigraphic and radiometric ages of both units, the reversed magnetizations probably represent Chron 33r [Swenson and McWilliams, 1989], which has a duration of only 4 m.y. [Harland et al., 1982; Kent and Gradstein, 1985]. Regardless of which scenario is correct, both dictate that an episode of rapid APW occurred at or before 80 Ma. Furthermore, the fact that the Maudlow Formation is surrounded and cut by thrust faults [Skipp and McGrew, 1977] and that areas adjacent to these rocks have experienced vertical-axis rotations, I would argue the paleomagnetic pole determined from the Elkhorn Mountains Volcanics best represents the position of the apparent pole at 80 Ma. Nevertheless, the results may be fortuitous because of the tectonic setting of the Elkhorn Mountains Volcanics and should be viewed with caution until more data are available for this interval of time.

Acknowledgments. S. D. Sheriff and C. A. Chesner assisted in the field. M. E. Beck, Jr., made the paleomagnetics lab at Western Washington University available to the author while on sabbatical leave, and R. Burmester made working in the lab a painless process. G. Johnson assisted in the lab. Constructive reviews by R. Butler and an anonymous reviewer greatly improved the manuscript. This work was supported with a grant (EAR-8803521) from the National Science Foundation.

21562202b, 1991, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/91JB00959 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

References

Cox, A., Anomalous remanent magnetization of basalt, *U. S. Geol. Surv. Bull., 1083E*, 131-160, 1961.

Cox, A., Confidence limits for the precision parameter *k*, *Geophys. J. R. Astron. Soc., 18*, 545-549, 1969.

Diehl, J. F., S. Beske-Diehl, M. E. Beck, Jr., and B. C. Hearn, Jr., Paleomagnetic results from Early Eocene intrusions, north-central Montana: Implications for North American apparent polar-wandering, *Geophys. Res. Lett., 7*, 541-544, 1980.

Diehl, J. F., M. E. Beck, Jr., S. Beske-Diehl, D. Jacobson, and B.C. Hearn, Paleomagnetism of the Late Cretaceous-Early Tertiary north-central Montana alkalic province, *J. Geophys. Res., 88*, 10593-10610, 1983.

Eldridge, S., and R. Van der Voo, Paleomagnetic study of thrust sheet rotations in the Helena and Wyoming salients of the northern Rocky Mountains, *Geol. Soc. Am. Mem., 171*, 319-332, 1988.

Engebretson, D. C., A. Cox, and R. G. Gordon, Relative motions between oceanic and continental Plates in the Pacific Ocean, *Geol. Soc. Am. Spec. Pap., 206*, 59 pp., 1985.

Fisher, R. A., Dispersion on a sphere, *Proc. R. Soc. London, Ser. A, 217*, 295-305, 1953.

Globerman, B. R. and E. Irving, Mid-Cretaceous paleomagnetic reference field for North America: Restudy of 100 Ma intrusive rocks from Arkansas, *J. Geophys. Res., 93*, 11,721-11,733, 1988.

Graham, K. W. T., The remagnetization of a surface outcrop by lightning currents, *Geophys. J. R. Astron. Soc., 6*, 65-102, 1961.

Gunderson, J. A., and S. D. Sheriff, A new Late Cretaceous paleomagnetic pole from the Adel Mountains, west-central Montana, *J. Geophys. Res., 96*, 317-326, 1991.

Hagstrum, J. T., and P. B. Gans, Paleomagnetism of the Oligocene Kalamazoo Tuff: Implications for Middle Tertiary extension in east central Nevada, *J. Geophys. Res., 94*, 1827-1842, 1989.

Hanna, W. F., Paleomagnetism of Upper Cretaceous rocks of southwestern Montana, *J. Geophys. Res., 72*, 595-610, 1967.

Harland, W. B., A. V. Cox, P. G. Llewellyn, C. A. G. Pickton, A. G. Smith, and R. Walters, *A Geologic Time Scale*, 131 pp., Cambridge University Press, New York, 1982.

Kent, D. V., and F. M. Gradstein, A Cretaceous and Jurassic geochronology, *Geol. Soc. Am. Bull., 96*, 1419-1427, 1985.

Klepper, M. R., M. R. Weeks, and E. T. Ruppel, Geology of the southern Elkhorn Mountains, Jefferson and Broadwater counties, Montana, *U.S. Geol. Surv. Prof. Pap., 292*, 82 pp, 1957.

Klepper, M. R., E. T.. Ruppel, V. L.. Freeman, and R. A. Weeks, Geology and mineral deposits, east flank of the Elkhorn Mountains, Broadwater county, Montana, *U.S. Geol. Surv. Prof. Pap., 665*, 66 pp., 1971.

Lipman, P. W., The roots of ash flow calderas in western North America: Windows into tops of granitic batholiths, *J. Geophys. Res., 89*, 8801-8841, 1984.

MacDonald, W. D., Net tectonic rotation, apparent tectonic rotation, and the structural tilt correction in paleomagnetic studies, *J. Geophys. Res., 85*, 3659-3669, 1980.

McFadden, P. L. and M. W. McElhinny, The combined analysis of remagnetization circles and direct observations in palaeomagnetism, *Earth Planet. Sci. Lett., 87*, 161-172, 1988.

McFadden, P. L., R. T. Merrill, and M. W. McEhlinny, Dipole/quadrupole family modeling of paleosecular variation, *J. Geophys. Res., 93*, 11,583-11,588, 1988.

Robinson, G. D., and R. F. Marvin, Upper Cretaceous volcanic glass from western Montana, *Geol. Soc. Am. Bull., 78*, 601-608, 1967.

Robinson, G. D., M. R. Klepper, and J. D. Obradovich, Overlapping plutonism, volcanism, and tectonism in the Boulder batholith region, western Montana, *Geol. Soc. Am. Mem., 116*, 557-576, 1968.

Rosenbaum, J. G., Paleomagnetic directional dispersion produced by plastic deformation in a thick Miocene welded tuff, southern Nevada: Implications for welding temperatures, *J. Geophys. Res., 91*, 12,817-12,834, 1986.

Ruppel, E. T., C. A. Wallace, R. G. Schmidt, and D. A. Lopez, Preliminary interpretation of the thrust belt in southwest and west-central Montana and east-central Idaho, in *Southwest Montana: Montana Geologic Society Field Conference and Symposium Guidebook*, edited by T. E. Tucker, pp. 139-159, Billings, 1981.

Schmidt, R. G., Geologic map of the Coburn Mountain quadrangle, Lewis and Clark and Cascade counties, Montana, *U. S. Geol. Surv. Quad. Map, GQ-974*, 1972.

Skipp, B., and L. W. McGrew, The Maudlow and Sedan formations of the upper Cretaceous Livingston Group on the west edge of the Crazy Mountain Basin, Montana, *U. S. Geol. Surv. Bull., 1422-B*, 65 pp., 1977.

Skipp, B., and A. D. Peterson, Geologic map of the Maudlow quadrangle, southwestern Montana, *U. S. Geol. Surv. Misc. Geol. Invest. Map, I-452*, 1965.

Smedes, H. W., Geology and igneous petrology of the northern Elkhorn Mountains, Jefferson and Broadwater counties, Montana, *U.S. Geol. Surv. Prof. Pap., 510*, 116 pp., 1966.

Streeter, M. E., The geology of the southern Bull Mountain area, Jefferson County, Montana: M.S. thesis, Western Michigan University, Kalamazoo, 1983.

Swenson, P., and M. McWilliams, Paleomagnetic results from the Upper Cretaceous Maudlow and Livingston formations, southwest Montana, *Geophys. Res. Lett., 16*, 669-672, 1989.

Tilling, R. I., M. R. Klepper, and J. D. Obradovich, K-Ar ages and time span of emplacement of the Boulder batholith, Montana, *Am. J. Sci., 266*, 671-689, 1968.

Vugteveen, R. W., A. E. Barnes, and R. F. Butler, Paleomagnetism of the Roskruge and Gringo Gulch volcanics, southeast Arizona, *J. Geophys. Res., 86*, 4021-4028, 1981.

Weeks, R. A., Geologic map of the Bull Mountain area, Jefferson County, Montana, *U. S. Geol. Surv. Open File Rep., 74-354*, 1974.

21562202b, 1991, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/91JB00959 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Zijderveld, J. D. A., A. C. demagnetization of rocks: Analysis of results, in *Methods in Paleomagnetism*, edited by D. W. Collinson, K. M. Creer, and S. K. Runcorn, pp. 254-286, Elsevier, New York, 1967.

J. F. Diehl, Department of Geological Engineering, Geology, and Geophysics, Michigan Technological University, Houghton, MI 49931.

(Received February 18, 1991;
accepted April 2, 1991.)

21562202b, 1991, B6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/91JB00959 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License